

Jianxiang Zhao

Curriculum Vitae

Postdoctoral Fellow

Department of Mechanical & Industrial Engineering

University of Toronto

5 King's College Road, Toronto, ON, Canada

Email: jxzhao@mie.utoronto.ca

EDUCATION

PhD, Mechanical Engineering

University of Toronto, Canada | 2021–2025

MSc, Chemical Process Equipment

Zhejiang University, China | 2017–2020

BSc, Control Engineering

Xi'an Jiaotong University, China | 2013–2017

RESEARCH INTERESTS

Multifunctional polymeric materials; advanced manufacturing techniques; micro/nano-scale structures; porous structures; physics-informed machine learning; thermal management and passive cooling.

TECHNICAL EXPERTISE

Micro/Nano Structuring & Fabrication

Foaming of micro/nano cellular structures; micro/nano-layered film and foam structures; nanofibrillation (fiber-in-fiber) processing; porous microcapillary film extrusion; multilayer coextrusion; gas-assisted microextrusion; CO₂ and UV laser micro/nanostructuring (LIFT).

Functional Device & System Design

Triboelectric nanogenerator (TENG) fabrication; passive daytime radiative cooling materials; thermal management and insulation structures; EMI shielding composites; shape-memory polymer actuator systems; flexible and stretchable substrate processing.

PUBLICATIONS

First-authored Publications

1. **Zhao, J.**, et al. Temperature-adaptive radiative cooling for building envelopes via shape-memory-driven thermal conductivity switching. *Advanced Functional Materials*. To be submitted.
2. **Zhao, J.**, Al Faysal, A., Lee, J. U., Filleter, T., & Lee, P. C. Toughening mechanism in micro-layered polymer films revealed by in situ micro-tensile testing. *ACS Applied Materials & Interfaces*, 2026. Submitted.
3. **Zhao, J.**, Zhang, L., Lee, J. U., Sansone, N. D., Kim, J., Bazbaz, L., Al Faysal, A., & Lee, P. C. Butterfly wing microstructure inspired solid/porous alternating layered structures: In situ visualization of confined foaming. *Small* 21.40 (2025): e08472.
4. **Zhao, J.**, Li, H., Choi, D., & Lee, P. C. Micro-layered film and foam alternating structures: A novel passive

daytime radiative cooling material. *Nano Energy*, 2024, **127**, 109695.

5. **Zhao, J.**, Zheng, S., Xu, Z., & Huang, X. Microwave foaming of polymers. *Journal of Materials Science*, 2021, 15491–15498.
6. **Zhao, J.**, Xu, Z., Zheng, S., Liu, C., Liu, J., & Pei, H. Solvent-free generation of foamed microcapillary films with a dual network of interconnected pores and internal channels. *Macromolecular Materials and Engineering*, 2020, 1900768.

Co-authored Publications

6. Lee, J. U., Aguiar, R., Su, T. Y. T., Kim, K.-W., **Zhao, J.**, et al. Bio-inspired laser micro/nanostructuring of highly graphitic carbon enabling structural coloration and high-performance all-weather solar evaporator. *Chemical Engineering Journal*, 2026, 174935.
7. Lee, J. U., Aguiar, R., **Zhao, J.**, et al. In situ UV laser-induced forward transfer of carbon nanosphere/graphene composites on siloxane–polyurethane substrates for human health monitoring. *Advanced Composites and Hybrid Materials*, 2026.
8. Faysal, A. A., **Zhao, J.**, & Lee, P. C. Layered foam/film polymer composite structure exhibiting high absorption dominant electromagnetic interference (EMI) shielding properties. *Americas Regional Meeting of the Polymer Processing Society*, Guelph, Canada, 2025.
9. Faysal, A. A., Mallamaci, G., **Zhao, J.**, Sollogoub, C., & Lee, P. C. Multi-interphase biodegradable nanolayer films for enhanced mechanical and gas barrier properties. *Sustainable Materials and Technologies*, 2025. To be submitted.
10. Lee, T., Lee, J., **Zhao, J.**, & Lee, P. C. Artificial nacre materials made by CO₂ laser-induced technology for graphene structures with stacked PBAT films embedded with cellulose-rich biofibers (CRB) for enhanced material properties. *Advanced Functional Materials*, 2025. To be submitted.
11. *Image recognition enabled automatic in situ tracing of cell nucleation and growth in micro-layered film/foam alternating structures*. 2025. In preparation.
12. Wang, L., Huang, X., **Zhao, J.**, Liu, C., Xu, Z., & Pei, H. Production of high aspect-ratio rectangular-lumen microtube by gas-assisted microextrusion. *Journal of Applied Polymer Science*, 2020, 49352.
13. Huang, X., Wang, L., **Zhao, J.**, Liu, C., Zhou, F., Xu, Z., & Pei, H. Fabrication and application of flexible rectangular microtubes. *Polymer-Plastics Technology and Materials*, 2020, 959–965.
14. Wang, M., Kong, C., Liang, Q., **Zhao, J.**, Wen, M., Xu, Z., & Ruan, X. Numerical simulations of wall contact angle effects on droplet size during step emulsification. *RSC Advances*, 2018, 33042–33047.
15. T. Su, A. Al Faysal, **J. Zhao**, & P. C. Lee. Nano-Layered Films and Foams. *Polyester Films: Materials, Processes and Applications*, pp. 225–276.

Conference Presentations

16. Zhao, J., & Lee, P. C. Bio-mimetic micro-/nano-layered foams: In situ visualization of cell nucleation and growth dynamics. Society of Plastics Engineers, ANTEC 2025, Philadelphia, PA, USA, Mar 3–6, 2025.
17. Zhao, J., Li, H., & Lee, P. C. Micro/nano-layered film/foam structures: A novel approach to passive cooling building materials. Society of Plastics Engineers, ANTEC 2024, St. Louis, MO, USA, Mar 4–7, 2024.
18. Zhao, J., Li, H., & Lee, P. C. Micro/nano-layer structured foams with tunable foam morphology and properties. Society of Plastics Engineers, ANTEC 2023, Denver, CO, USA, Mar 27–30, 2023.
19. Zhao, J., & Lee, P. C. High thermal insulation structures using micro-/nano-layered film/foam technology. 37th International Conference of the Polymer Processing Society (PPS-37), Fukuoka, Japan, Apr 11–15,

2022.

PATENTS

19. A movable-block variable-differentiation-cross section extrusion die for preparing micro-nano structures. China. CN110253856A.

20. A method for generating foamed microcapillary films with a dual network of interconnected pores and internal channels. China. PCT/CN2019/096518.